order23 RESULTS
PREREG sha256 e390d61b9330c3e61e7bb10f2b9d8c8b6b7f21cf628d14c2b4d341d1429a2f03 (hashed before numerics; RG9b-corrected, RG10, RG11 hashed there).
test	value	verdict
C1 corner formula = CO (x and −1/x)	0.0	PASS
C2 Γ_σ·G = G	0.0	PASS
RG9b corrected (vertex ¼·[α∈Z²]; 2 such companions)	max abs 7.1e−15	**PASS**
RG11 Γ_σ·CO = Corner_{σK}	max abs 7.1e−15; displacement 0.0	**vacuous** (Γ_σ = I exactly)
RG10 U = Corner_K + Corner_{σK}, law U(−1/x) = x·kM·U(x)	abs 108.8 (‖S‖ = 1856), rel 0.0586; fitted κ gives 0.0585	**FAIL**
Reading
Γ_σ is the identity on the 105 companions (σ preserves each, as in Prop. Bcp), so Corner_{σK} = Corner_K and U = 2·A2_corner: the σ-sum adds nothing and RG10 reduces to the order19 law (same relative residual). The geometric near-law does not hold in this form.
Category A (folded into v8)
Unimodular-cone corner formula; subdivision with vertex term; transport Γ_g·Cor_K = Cor_{g⁻¹K} for automorphs; Γ_σ = I ⇒ Cor_{σK} = Cor_K.
Paper
RTA_FOUNDATION_BOUND_v8.pdf, 133 pages.

order23 NOTES
- RG10 was non-vacuous by its preregistered norm test (‖U‖ = 2‖CO‖), but that test did not anticipate Γ_σ = I; in substance RG10 repeats order19 L0. Disclosed.
- RG11 "vacuous" by rule: the reflection fixing ĉ′ acts trivially on the companions, so it cannot supply a new geometric law.
- Next geometric candidate (to preregister in order24): use the γ-orbit instead of σ, e.g. the γ-average over one period of cones, Σ_{n} Cor_{γ^{-n}K} truncated symmetrically, and test whether γ-symmetrisation removes the P-even law defect on x>0; or the difference Cor_{γ⁻¹K} − Cor_K (the γ-defect) as the object to test against the w=1 law.
- Lock: MAPS_CANDIDATES untouched; no Resonant objects; no bridge claim. E1–E3 OPEN. Ξ not identified.
- nice, OMP_NUM_THREADS=1, <3 GB, ~35 s; no push/sends; new files only.